

Week 10, November 13, 2024

Main topics

- Nonlinear inverse problems
- Linearization

Material

From last week: Kirsch 6.1-6.3 (EIT)

New material: Kirsch 4-4.1, Appendix A.7

Program

08:00-08:30 Local ill-posedness; EIT: lack of stability in the concentric case.

08:30-10:30 Exercises (see below)

10:45-12:00 Lecture: Nonlinear inverse problems and linearization

Exercises

1. Consider the autoconvolution operator K in $L^2(0, 1)$ defined by

$$K(x)(t) = \int_0^t x(t-s)x(s) \, ds, \quad t \in (0, 1).$$

- (a) Show that $K(x) \in L^2(0, 1)$ for $x \in L^2(0, 1)$ and satisfies

$$\|K(x)\|_{L^2(0,1)} \leq \|x\|_{L^2(0,1)}^2$$

- (b) Adapt your implementation of the Abel problem (week 1) to approximate the autoconvolution operator in the following way: Write a function that takes a convolution kernel x and gives a matrix $A(x)$. Then $K(x)$ should be approximated by $A(x) * x$. Experiment with your favourite functions x .
- (c) Consider the uniqueness question: Show that $K(x) = 0$ implies $x = 0$.
Hint: Extend any function $z \in L^2(0, 1)$ by zero outside $(0, 1)$ to $\tilde{z} \in L^2(\mathbb{R})$; use Fourier transform and convolution property $\widehat{\tilde{y}} = |\widehat{\tilde{x}}|^2$ to draw the conclusion ($\tilde{x}, \tilde{y}$ being the extensions of x, y respectively).
- (d) Use the fact that $K(\alpha x) = \alpha^2 K(x)$ to construct a counter-example to uniqueness. Is this consistent with the above?

(e) Fix an element $x^\dagger \in L^2(0, 1)$ and consider for each $n \in \mathbb{N}$ the function

$$x_n(s) = \begin{cases} x^\dagger(s), & 0 < s \leq 1 - \frac{1}{n}, \\ x^\dagger(s) + r\sqrt{n}, & 1 - \frac{1}{n} < s < 1. \end{cases}$$

Show that $\|x_n - x^\dagger\|_{L^2(0,1)} = r$ while

$$\|K(x_n) - K(x^\dagger)\|_{L^2(0,1)} \leq \frac{2}{\sqrt{n}} \|x^\dagger\|_{L^2(0,1)} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Hint: Use the Minkowski inequality in integral form on the way.

What is the conclusion regarding local ill-posedness for the inverse problem?

2. Suppose $\Omega = \mathbb{D} \subset \mathbb{R}^2$ and let

$$\gamma(x) = \begin{cases} 1 + \hat{q}, & |x| \leq R, \\ 1, & R < |x| \leq 1. \end{cases}$$

When the boundary current $g_n(\theta) = e^{in\theta}$, $n \neq 0$, is injected, the measured boundary voltage is $f_n(\theta) = \lambda_n e^{in\theta}$ with

$$\lambda_n = \frac{1}{|n|} \frac{\alpha - R^{2|n|}}{\alpha + R^{2|n|}}, \quad \alpha = 1 + \frac{2}{\hat{q}}.$$

That is, the current to voltage map is diagonal in the Fourier basis:

$$\Lambda_\gamma e^{in\theta} = \lambda_n e^{in\theta}.$$

The inverse problem is then reduced to: given eigenvalues $\{\lambda_n\}$ find $\hat{q}$ and R .

- (a) Set up a numerical code that given the contrast $\hat{q}$ and radius R , compute λ_n .
- (b) Suppose we are given eigenvalues $\lambda_1 = 0.914$, $\lambda_2 = 0.496$ (perturbed by a little noise). Find $\hat{q}$, R . (You might want to use *fsolve* or similar black box numerical solver.)